MIL-I-5154B(USAF)
16 January 1968
Superseding
MIL-I-5154A(USAF)
25 June 1952

MILITARY SPECIFICATION

INDICATOR, TRUE AIRSPEED, TYPE M-5

1. SCOPE

1.1 This specification covers one type of pitot static true airspeed indicator, designated Type M-5.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawing, Engineering and Associated List
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-726	Packaging Requirements Code
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-834	Packaging Data Forms Instructions for Preparation and Use of
MS24693	Screw, Machine, Flat Countersunk Head, 100 Degree, Cross Recessed, UNC-2A and UNF-2A
MS28105	Window, Dial-Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33638	Cases, Instrument, Flange-Mounted, Aircraft Case, Instrument, Flange-Mounted, Aircraft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. The true airspeed indicator furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

*3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Metals. Metals shall be of the corrosion resistant type or shall be suitably protected to resist corrosion during normal service life.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.3.3 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design and construction.

3.4.1 General design. The indicator shall be designed to indicate true airspeed at any altitude from 1000 feet below sea level to 50,000 feet above sea level and under free air temperature conditions from +40 to -60°C.

3.4.2 General construction. The indicator shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2.1 Adjustment and repairs. The indicator shall be constructed so that adjustment and repairs can be easily made by the personnel of operating units and overhaul bases.

3.4.3 Case. The case shall be as shown on Figure 1 and shall be made either of nonferrous low density metal or of synthetic material uniform in texture with a smooth surface and entirely covered with a durable black finish. Synthetic material shall be of the thermal setting type composed of a suitable filler and phenol condensation product binder.

3.4.3.1 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator. If a synthetic material is used in the construction of the case, one brass or aluminum alloy insert for the static pressure connection, one for the pitot pressure connection, and one for the free air temperature capillary tubing connection shall be molded in the base or suitably fastened by screws. If of aluminum alloy, the insert shall be anodically treated except for the threads.

3.4.3.2 Bezel ring. The bezel ring shall be made of nonferrous low density metal and shall have a durably dull black finish. The ring shall be held in place by means of screws properly secured by lock washers or similar devices of approved design. Provision shall be made for replacement of the cover glass by removal of the bezel ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

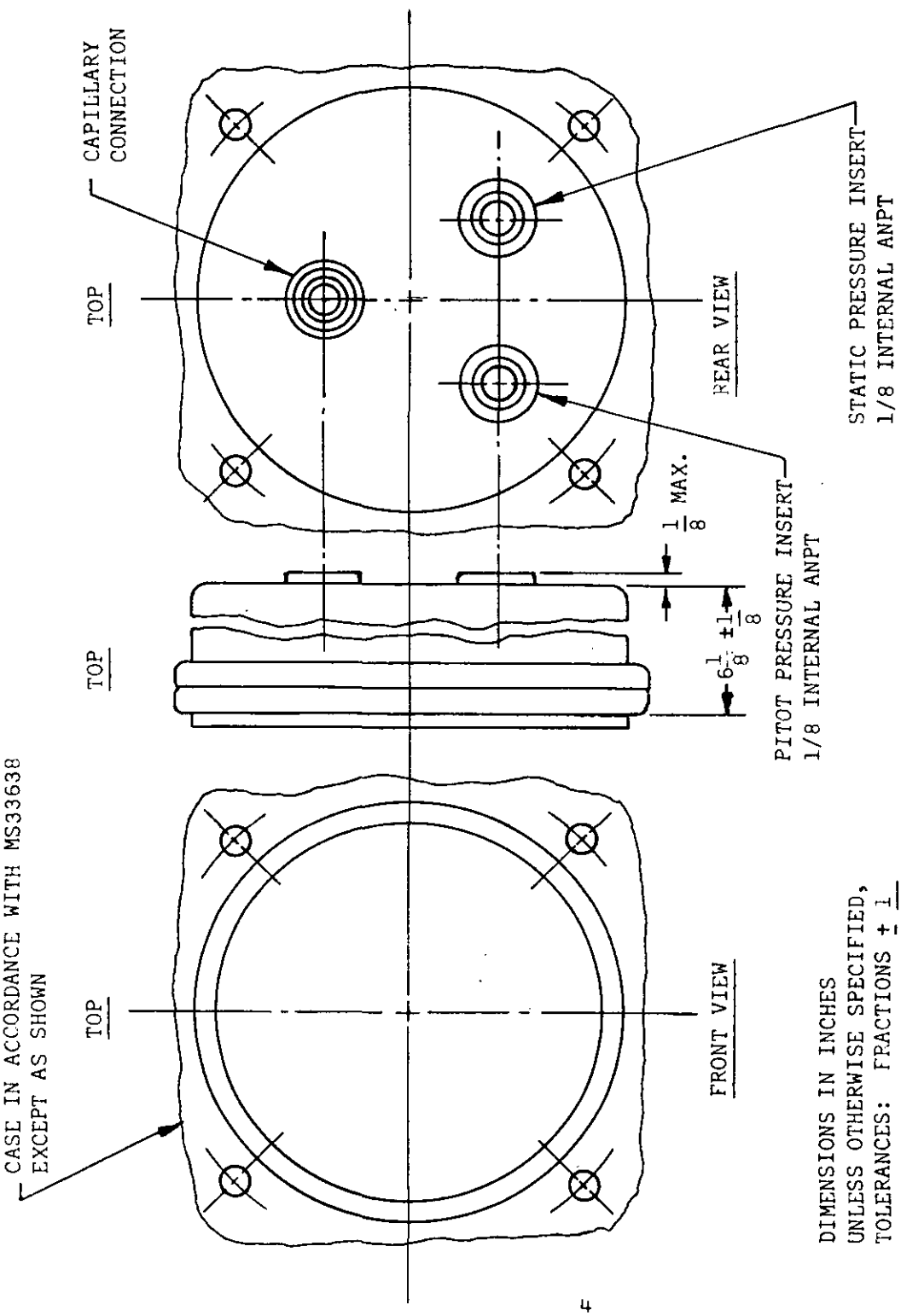


FIGURE 1. Case

Source: https://assist.dla.mil -- Downloaded: 20261-17T04:23Z
Check the source to verify that this is the current version before use.

3.4.3.3 The back of the case adjacent to the pressure connections shall be marked as follows:

"P" Pressure Connection
"S" Static Connection

3.4.3.4 The following shall be marked on the back of the case:

CAUTION
DO NOT BLOW IN TUBES

3.4.4 Cover glass. The cover glass shall be in accordance with MS28105 and shall be properly sealed. The distance between the cover glass and the dial shall be the minimum practicable and shall not exceed 0.187 inch.

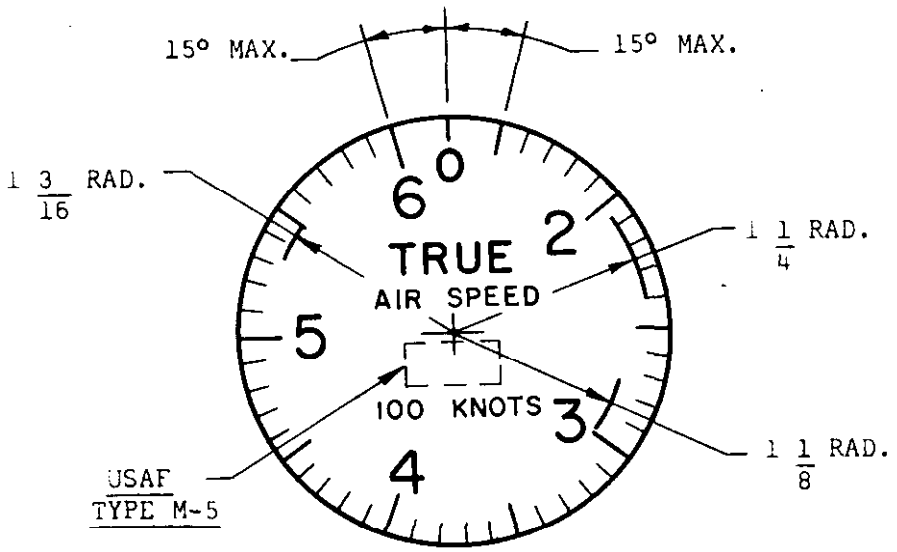
3.4.5 Instrument dial. The instrument dial shall conform to Figure 2. The style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.4.5.1 Dial markings.

3.4.5.1.1 Fluorescent-luminescent material. The following markings shall be finished in fluorescent-radioactive luminescent material in accordance with MIL-L-25142. The dimensions of the markings shall be as follows:

	Height or Length Inch + 1/64	Width of Line or Graduation Inch + 0.005
Numerals 2 to 6 as shown on Figure 2	5/16	0.031
100-knot graduations	(Figure 2)	0.031
50-knot graduations	(Figure 2)	0.031
10-knot graduations from 150 to 600 knots inclusive as indicated on Figure 2	(Figure 2)	0.015
Lettering "TRUE"	3/16	0.031
Zero numeral	5/32	0.015
Zero graduation	1/8	0.015
Lettering "100 KNOTS"	1/8	0.015
Lettering "AIRSPEED"	1/8	0.015

Source: https://assist.dla.mil -- Downloaded: 20261O-17T04:23Z
Check the source to verify that this is the current version before use.



DIAL GRADUATIONS LINEAR
FROM 150 TO 600 KNOTS
ALL GRADUATIONS ARE
PARTIAL RADIAL LINES

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCE ON:
-ANGLES-
-DECIMALS-
-FRACTIONS- ± 1/64

FIGURE 2. Dial

Source: https://assist.dla.mil -- Downloaded: 20261 0-17T04:23Z
Check the source to verify that this is the current version before use.

3.4.5.1.2 Durable dull black. The markings "USAF" and "TYPE M-5" shall be durably and legibly marked on the dial in durable dull black lettering 3/64 inch high. These markings and all other markings not otherwise specified shall be finished in durable dull black.

3.4.5.1.2.1 The background of the dial, the unshaded part of the pointer and the front of the bezel ring shall be finished in durable dull black.

3.4.5.2 Visibility of dial. The pointer, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.4.6 Pointer. The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. It shall be in accordance with MS33585, except that the shaded portion of the pointer shall be finished in luminous material in accordance with MIL-L-25142. The pointer shall be firmly attached to the mechanism and yet be readily adjustable.

3.4.6.1 Limitation of pointer movement. The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 359 degrees of arc clockwise from the zero graduation when the indicator is submitted to overpressure tests. Suitable stops may also be incorporated in the indicator mechanism to limit the counterclockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity.

3.4.7 Threads.

3.4.7.1 Screw. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.4.7.2 Pipe. Pipe threads shall conform to MIL-P-7105.

3.4.8 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.4.9 Temperature and pressure compensation. The indicator shall be designed so that proper compensation is made for variation in pressure and temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.4.10 Lubrication. Where lubrication is necessary, the lubricant shall be of a quality such that satisfactory lubrication will be provided when the indicator is operated at temperatures in the range of +71 to -54°C. The lubricating oil used shall be approved by the procuring activity.

3.4.11 Temperature bulb. The free air temperature bulb shall be designed for mounting flush with the skin of the airplane and shall be in accordance with Figure 3. Screws, nuts, washers, gaskets, et cetera, for installing the temperature bulb shall be furnished. The gaskets shall be furnished in such thickness that a sealed installation may be obtained on pressurized cabin aircraft with skin thickness from 1/32 to 3/16 inch thick.

3.4.12 Capillary tubing. The free air temperature capillary tubing shall be covered throughout its length with a flexible braided casing. The casing shall be securely attached to the case and to the bulb to protect the capillary tubing against strains.

3.4.12.1 Location. The location of the connection of the free air temperature capillary tube to the indicator case shall conform to Figure 1.

3.4.12.2 Length. The length of the free air temperature capillary shall be 8 feet, unless otherwise specified by the procuring activity. The capillary length shall be marked on the case.

3.4.12.3 Capillary reinforcement. A suitable flexible reinforcement extending 4 inches from the case and from the bulb shall be provided in each end of the free air temperature capillary casing. The reinforcement shall be of a flexibility to withstand, without failure, a 180-degree bend of 1-3/8 inch radius. The reinforcement shall be securely attached to the case and to the bulb. The inside diameter of the reinforcement shall be such that the reinforcement will slide freely on the capillary casing. The reinforcement shall be braided brass or bronze spiral armor approximately 9/32 inch outside diameter, plain weave, 7 strands. The brass or bronze wire shall be 0.011 inch in diameter with 6-1/2 to 7 strands per inch.

3.4.12.4 Cleaning. The free air temperature capillary tubing shall be thoroughly cleaned at the proper stage of assembly and shall be filled in a manner to exclude air and other noncondensing gases and moisture.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

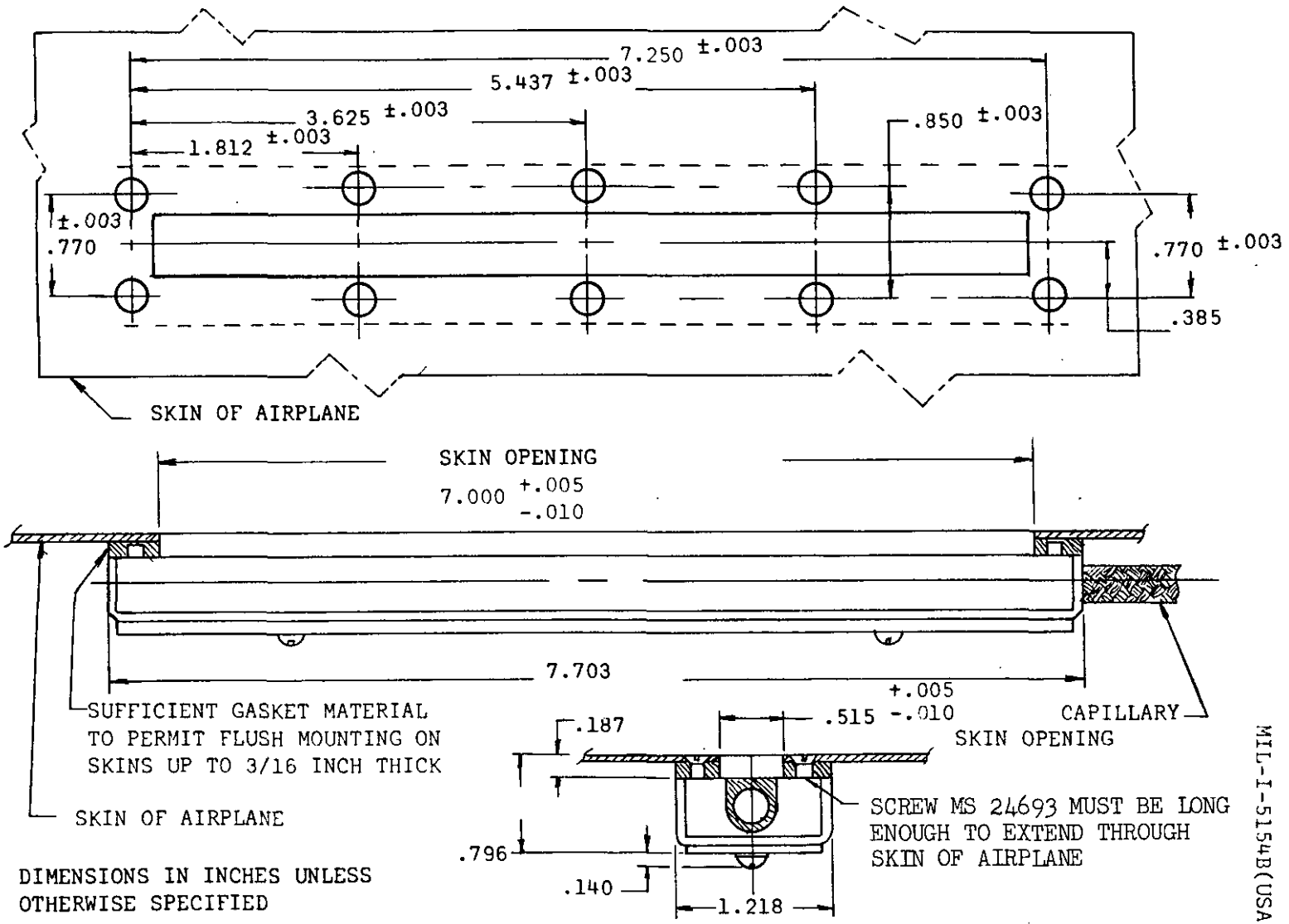


Figure 3. Temperature Bulb

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.6 Weight. The weight of the indicator completely assembled, including mounting screws, shall not exceed 4.0 pounds.

3.7 Finish and protective coating.

3.7.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, and case inserts need not be anodized.

3.7.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416 Type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

*3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9 Installation.

3.9.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the indicator. They shall be No. 6-32 NC-2A round head brass machine screws having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 3/16 inch thick.

3.9.2 Envelope. An envelope containing the mounting screws shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING
SCREWS AND ACCESSORIES

3.9.3 Accessories. The contractor shall furnish with each indicator a sun shield, gaskets, and sufficient mounting screws and nuts for installation of the flush mounted temperature bulb.

3.10 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies; and alignment of parts and tightness of assembly screws.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

*4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a.	Qualification tests	See 4.3
b.	Acceptance tests	See 4.4

*4.3 Qualification tests. The qualification tests shall consist of all the tests of this specification.

*4.3.1 Qualification test samples. Qualification tests shall be performed on three indicators. These samples shall be idenfified as qualification samples and forwarded to the preparing activity.

*4.4 Acceptance tests. Acceptance tests shall consist of individual tests and sampling tests.

*4.4.1 Individual tests. Each indicator submitted for acceptance under contract shall be subjected to the following tests as described in 4.6:

a.	Examination of product	See 4.6.1
b.	Scale error	See 4.6.5
c.	Friction	See 4.6.6
d.	Leakage	See 4.6.7
e.	Inspection and test	See 4.8

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the currenversion before use.

4.4.2 Sampling tests. Five indicators shall be selected at random from each lot of 500 or fraction thereof and subjected to the following tests as described in 4.6:

a.	Weight	See 4.6.8
b.	Magnetic property	See 4.6.9
c.	Low temperature	See 4.6.10
d.	High temperature	See 4.6.11
e.	Seasoning	See 4.6.12
f.	Capillary temperature	See 4.6.13
g.	Capillary bending and strain	See 4.6.14
h.	Vibration error	See 4.6.15
i.	Overpressure	See 4.6.16
j.	Position error	See 4.6.17

*4.4.2.1 Rejection and retest. When one or more indicators from a lot fail to pass one or more of the sampling tests, acceptance of the lot will be withheld until the extent and cause of failure are determined. Rejected indicators shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

*4.4.2.2 Individual test may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the lot will not be made, however, until it is determined that the lot meets all the requirements of the specification.

*4.4.2.3 Defects in items already accepted. The investigation of a sampling test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

*4.5.1 Standard conditions. Unless otherwise specified, all tests are to be conducted at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 77°F. When tests are conducted at conditions substantially different from the above values, proper allowances shall be made for the differences which could change instrument readings or indicator performance.

4.5.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.5.3 Attitude. Unless otherwise specified, the indicator shall be tested in normal operating position.

4.5.4 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3000 cycles per second and shall subject the indicator to vibration such that a point on the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.5.5 Tolerances. Tolerances shall be as specified in Tables I through VII and in the applicable paragraphs.

TABLE I

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
+40	0	150	156
		200*	207
		250	258
		300*	308
	5000	150	170
		200*	226
		250	281
		300*	335

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

TABLE II

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
+20	0	150	151
		200*	200
		250	249
		300*	298
	5000	150	165
		200*	219
		250	272
		300*	324
	10,000	150	181
		200*	239
		250	297
		300*	353

TABLE III

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
0	0	200*	193
		250	241
		300*	287
	5000	150	159
		200*	211
		250	262
		300*	312
	10,000	150	175
		200*	231
		250	287
		300*	340
	15,000	150	192
		200*	254
		250	314
		300*	372

Source: https://assist.dla.mil -- Downloaded: 20260-17T04:23Z
Check the source to verify that this is the current version before use.

TABLE IV

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
-20	0	200*	186
		250	232
		300*	277
	10,000	150	168
		200*	223
		250	276
		300*	328
	15,000	150	185
		200*	244
		250	302
		300*	358
	20,000	150	204
		200*	269
		250	331
		300*	391

TABLE V

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
-40	0	200*	179
		250	222
		300*	265
	10,000	150	161
		200*	213
		250	265
		300*	315
	20,000	150	196
		200*	258
		250	318
		300*	375
	35,000	150	268
		200*	349
		250	423
		300*	491
	50,000	150	369
		200*	468

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

TABLE VI

C° Test Setting	Pressure Altitude Feet	Indicated Airspeed Knots	True Airspeed Knots
-60	0	200*	171
		250	213
		300*	254
	10,000	150	154
		200*	204
		250	253
		300*	301
	20,000	150*	187
		200	247
		250*	304
	35,000	150*	256

NOTE: Asterisks will refer reader to 4.6.5.3.4 and 4.6.6.

TABLE VII

Tests	Tolerances Knots
Friction	8.0
Position error	5.0
Vibration error	5.0
Vibration failure	5.0
Scale error	
Seasoning	5.0
Low temperature operation	5.0
High temperature operation	5.0
Overpressure	5.0
Capillary temperature	2.0
Capillary bending and strain	2.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

True airspeed knots in the tables of this specification are calculated from the following equation:

$$V \text{ true} = 87.0114 \sqrt{T_o \left[\left(1 + \frac{dP}{P}\right)^{.2857} - 1 \right]}$$

Where: V_{true} = true airspeed in knots
T_o = corrected temperature in degrees K (free air temperature)
T_o = $(273.2 + T) - [1.07(V \text{ true})^2 \times 10^{-4}]$
Where T is test temperature
P = pressure corresponding to given altitude
dP = pressure corresponding to given indicated airspeed (same units as P)

4.6 Test methods.

*4.6.1 Examination of product. Each indicator submitted for acceptance under contract shall be examined to determine conformance to this specification with respect to workmanship, material, design and construction, dimensions, protective coating and marking.

4.6.2 Environmental tests. Environmental tests shall be conducted in accordance with the specified procedures of MIL-E-5272 and as specified herein.

*4.6.2.1 Low temperature. Procedure II. While still at -54°C, pressure shall be applied to the pitot connection. The indicator shall operate without sticking. After the indicator has returned to room temperature, it shall meet the requirements specified for room temperature scale error. There shall be no damage which would adversely affect subsequent operation.

4.6.2.2 High temperature. Procedure II for a period of 24 hours. After the indicator has returned to room temperature, it shall meet the requirements specified for room temperature scale error. There shall be no damage which would adversely affect subsequent operation.

4.6.2.3 Temperature-altitude. Procedure II, except that the chamber shall be maintained at an altitude of 60,000 feet and at an ambient temperature of -54°C for a continuous period of four hours. There shall be no damage as a result of this test.

4.6.2.4 Vibration. Procedure V. The test shall be conducted while a differential pressure of 300 knots is applied to the indicator. The difference between the indication of the pointer at the beginning and at the end of the three hour vibration period shall not exceed the tolerance specified in Table VII, and no looseness in the mechanism nor damage shall result in any part of the indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

4.6.2.5 Fungus. Procedure I. The indicator shall be complete with all applicable external connections. After the test, the indicator shall meet the requirements specified for individual tests. There shall be no deterioration which would adversely affect subsequent operation.

4.6.2.6 Salt spray. Procedure I for a period of 50 hours. At the end of the 50-hour period, the individual tests shall be repeated. No damage shall result from this test and the tolerances specified for the individual tests shall apply.

4.6.2.7 Humidity. Procedure I. The indicator shall be complete with all applicable external connections and shall be mounted in its normal operating position. There shall be no evidence of corrosion or rusting of the internal or external parts of the indicator.

4.6.3 Acceleration. The indicator shall be mounted and accelerated first on its vertical axis and then on each of two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20 g in each of the above axes for a period of 30 seconds. At the end of this acceleration, the indicator shall meet the requirements specified for room temperature scale error, except that only Table II shall be used.

4.6.4 Mounting lugs. The indicator case with the mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug along the lug axis and toward the rear of the instrument. The lug shall withstand the applied load without fracture.

4.6.5 Scale error. The indicator shall be mounted on a panel that has facilities for tapping to remove the friction effect. The free air temperature bulb shall be placed in a free air temperature test fixture in which it is possible to control, vary, and to read the temperature accurately throughout the range from -60 to +40°C.

4.6.5.1 The static pressure connection of the indicator shall be connected by means of a cross to an accurate barometer, a source of air pressure and suction, and to the suction side of a manometer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

4.6.5.2 The pitot pressure connection of the indicator shall be connected by means of a tee to a source of air pressure and to the pressure side of the manometer.

4.6.5.3 Each indicator shall be tested for scale errors as follows:

4.6.5.3.1 The temperature around the free air temperature bulb in the test fixture shall be adjusted to the test setting specified in Table I.

4.6.5.3.2 The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude as specified in Table I.

4.6.5.3.3 The pitot pressure shall then be adjusted so that the differential pressure indicated by the manometer is equal to the test setting as specified in Table I.

4.6.5.3.4 While the temperature of the free air temperature bulb is maintained at the test setting as specified in Table I, and the static pressure adjusted to zero feet altitude, the pitot pressure shall be varied so that the differential pressure is equal to each of the values marked by an asterisk listed in column 3 of Table I. Under these conditions, the true airspeed indicator shall indicate the true airpseeds listed in column 4 and shall not vary from these indications by more than the tolerance listed below. This procedure shall be followed for each of the different pressure altitudes listed in Table I.

a. 100 percent of the total recorded test points shall be within a tolerance of ± 10 knots.

b. 95 percent of the total recorded test points shall be within a tolerance of ± 8 knots.

c. 75 percent of the total recorded test points shall be within a tolerance of ± 5 knots.

4.6.5.3.5 The above procedure shall be repeated for the conditions specified in Tables II, III, IV, V, and VI.

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the current version before use.

4.6.5.3.6 The scale error test shall be made by subjecting the indicator to the differential pressures specified to produce the readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to give the desired reading; and with pressures decreasing, the pressure shall be brought down to, but shall not fall below the pressure specified to give the desired reading. The errors at the test points shall not exceed the tolerances specified in 4.6.5.3.4.

4.6.5.3.7 When desired by the Inspector, the indicator may be tested for scale error at any additional point or points on the scale, in combination with an indicated airspeed listed in the tables, and any pressure altitude and free air temperature. The errors at these additional test points shall not exceed the tolerances specified in 4.6.5.3.4.

4.6.6 Friction. The indicator shall be tested for friction at every point indicated by an asterisk in Tables I to VI, inclusive. The pressure shall be increased to bring the pointer approximately to the desired reading and then held constant while two readings are taken; the first before the indicator is tapped, and the second after the indicator is tapped. The difference between any two such readings shall not exceed the tolerance specified in Table VII. The pointer shall move smoothly, while the pressure is varied uniformly, without vibration of the indicator. This test may be combined with the test for scale errors.

4.6.7 Leakage.

4.6.7.1 Case leak. The pitot and static pressure connections of the indicator shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be applied to the pitot and static pressure connections of the indicator. With source disconnected during a period of 10 seconds, the difference between the mercury level in the manometer shall not change by more than 0.2 inch.

4.6.7.2 Diaphragm capsule leak. The static connection shall be opened to the atmosphere and a pressure sufficient to produce approximately full scale deflection of the pointer shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of one minute, the pointer shall not change its position by more than one knot.

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

4.6.8 Weight. The indicator shall be weighed completely assembled with mounting screws (see 3.6).

4.6.9 Magnetic property. The magnetic property of the indicator shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.18 ± 0.001 gauss, when the indicator is held in various positions on an east-west line five inches from the center of the magnet. An aircraft compass, with compensating magnets removed therefrom, may be used as the free magnet for this test. The maximum deflection of the magnet shall not exceed one degree.

4.6.10 Low temperature operation. The indicator shall be subjected to a temperature of -54 ± 2°C for a period of four hours. At the end of this period and while still at the low temperature, the indicator shall be tested for scale errors as specified in 4.6.5, except that only Tables V and VI shall be used. The error at the test points shall not exceed the tolerances specified below:

a. 100 percent of the total recorded test points shall be within a tolerance of ± 15 knots.

b. 95 percent of the total recorded test points shall be within a tolerance of ± 10 knots.

c. 75 percent of the total recorded test points shall be within a tolerance of ± 6 knots.

4.6.11 High temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of 71 ± 2°C for a period of four hours. While the indicator is still at this temperature it shall meet the requirements specified for scale errors as specified in 4.6.5, except that only Tables I and II shall be used. The errors at the test points shall not exceed the tolerances specified in 4.6.10.

4.6.12 Seasoning. The indicator shall be subjected to 1000 applications of a differential pressure sufficient to produce approximately full scale deflection. These application shall be made at a rate not greater than 6 per minute. Not less than one hour following this test, the indicator shall be tested for scale errors as specified in 4.6.5 at +20 ± 2°C and 5000 feet pressure altitude. The errors at the test points shall not exceed the tolerances in 4.6.5.3.4 by more than the tolerance specified in Table VII.

4.6.13 Capillary temperature. A three foot length of capillary shall be subjected to a temperature of 50°C with the indicator, the bulb, and the remainder of the capillary at room temperature. The change in reading due to heating the capillary shall not exceed the tolerance specified in Table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

4.6.14 Capillary bending and strain. The indicator reading shall be obtained for a bulb temperature of +50°C. The capillary shall then be bent 20 times in one place on a radius of 1/2 inch through an angle of 90 degrees and straightened out after each bending. The flexible braided capillary covering, at points approximately six inches from the case and from the bulb, shall be secured in a vise so that the case and the bulb are suspended vertically downward. Care shall be taken to secure the flexible covering lightly in the vise to avoid crushing the covering. The change in reading as the result of bending and straining the capillary shall not exceed the tolerance specified in Table VII.

4.6.15 Vibration error. Vibration error shall be in accordance with Procedure IV of MIL-E-5272. This test shall be conducted while differential pressures of 200 and 400 knots are applied to the indicator. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch. The maximum amplitude of pointer vibration and the pointer variation from its original position shall not exceed the tolerance specified in Table VII while the indicator is being vibrated.

4.6.16 Overpressure.

4.6.16.1 A pressure of 30 inches Hg shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes.

4.6.16.2 A pressure of 6 inches of water shall be applied to the static connection of the indicator for a period of 10 minutes.

4.6.16.3 Not less than five minutes following the application of the overpressure tests, the indicator shall be tested for scale error as specified in the scale error test at +20°C and 5000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in 4.6.5.3.4 by more than the tolerances specified in Table VII.

4.6.17 Position error. The indicator shall be observed while the indicator is held in any desired position and tapped. This reading shall not differ from the reading when the indicator is held in any other position by more than the tolerances specified in Table VII. This test shall be made at every other one of the points indicated by an asterisk in the tables beginning with Table I.

4.7 All parts, specimens, or assemblies destroyed in making tests required by this specification to determine compliance with the specification shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the currentversion before use.

4.8 Inspection and test. Testing of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Each indicator shall be preserved and packaged in accordance with Level "A" requirements as defined in MIL-STD-794 unless otherwise specified in the contract or purchase order. When detailed preservation and packaging instructions are not provided in the contract or purchase order, the appropriate preservation method will be determined in accordance with preservation selection chart, contained in MIL-STD-834. Accomplishment of selected methods shall be in accordance with MIL-P-116.

*5.2 Packing. Packing for shipment shall be in accordance with Level "B" requirements as defined in MIL-STD-794, unless otherwise specified in the contract or purchase order by the procuring activity.

*5.3 Marking of shipments. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Indicator, True Airspeed, One Scale 150 to 600 Knot Range, Type M-5 (Manufacturer's part no. *), Specification MIL-I-5154B.

*Applicable data to be entered by the contractor.

6. NOTES

6.1 Use. The Type M-5 true airspeed indicator is intended for use on very heavy bombardment aircraft to indicate true speed relative to the air.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.3 Definitions.

6.3.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the currenversion before use.

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Materiel Area (OCNE) Tinker AFB, Oklahoma and information pertaining to qualification of products may be obtained from that activity.

*6.5 Indication of change. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 71

Preparing activity:
Air Force -71

Review activity:
Air Force - 71

Project Number 6610-F009

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the currentversion before use.